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System saves $ for thrifty night owls

CAMBRIDGE, MA — Consumers may soon play a greater role in determining what they pay for electrical power.

The reason? Increased use by utility companies of time-of-day rates, which, like those used by phone companies, charge more during the day when usage is high than at night.

Time-of-day rates will enable consumers to save money by putting off activities, such as using washing machines, driers and water heaters, until evenings or weekends.

"A small life-style change may be necessary to benefit from time-of-day rating but we feel that people will make that change, because it will reduce their electrical costs," said Ralph Abbott, VP of American Science & Engineering, here. AS&E developed a two-way communications system — ASEP — that allows utility companies to implement time-of-day rates. The system is controlled by a Data General Eclipse S-230 minicomputer with 256K core memory.

Utilities in about 12 states now use such systems on experimental bases, and the Department of Energy estimates that most states will be using time-of-day rates within five or six years.

"Time-of-day rating is gaining increased acceptance because it benefits the consumer, allows the utility company to level off the power it must generate and, because it reduces the amount of oil needed to provide that power, it also aids our national goal of energy conservation," said Dr. Martin Annis, president of AS&E.

One of the first utilities to use the ASEP systems for time-of-day metering, the Florida Power Corp., will soon begin testing a dynamic rate system that will charge customers peak rates only when peak levels are actually reached. The ASEP system allows individual meter usage to be checked as often as every 30 minutes or as infrequently as once each month.

Because it is a two-way system, ASEP prevents electrical brownouts and blackouts and it operates over powerlines, a communications medium already in place.

"If electrical usage reaches dangerous levels," said Abbott, "the system is capable of turning off blocks of non-essential electrical loads. By doing this a major problem could be averted, with customers suffering only a minor inconvenience."

Florida Power and the Florida Power and Light Co. also plan to use the system to control residential air conditioning and heating systems and electrical water heaters. Wisconsin Electric Power uses ASEP to control water heaters in 154,000 homes in addition to remotely metering certain customers. Utility companies in Missouri, New Jersey, Minnesota and California have also implemented the system.